# Zen Legal: AI-Assisted Legal Research and Analysis

Technical Report v2025.10

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October 2025

**Abstract**

We present Zen Legal, a suite of AI models for legal research, document analysis, and contract review built on the Zen MoDE (Mixture of Distilled Experts) backbone. Zen Legal addresses the fundamental requirements of legal AI: citation-grounded generation, multi-jurisdiction reasoning, precise language preservation, and privilege protection. On standardized legal benchmarks: LegalBench 81.4%, CUAD contract clause extraction 89.3%, contract clause classification F1 0.923, and case outcome prediction 74.2%. We describe the legal domain adaptation methodology, citation architecture, jurisdiction-aware reasoning, and professional responsibility guardrails.

## 1 Introduction

Legal practice demands precision at a level that distinguishes it from most AI application domains. A misquoted statute, a missed precedent, or a hallucinated citation can undermine a legal argument, expose a client to liability, or trigger bar discipline for the attorney who relied on it.

Zen Legal is built on the premise that the primary role of AI in legal contexts is *augmentation*, not replacement—reducing research time from days to hours, identifying overlooked precedents, flagging inconsistencies in contract drafts, and generating first drafts that attorneys review and refine. This framing shapes every design decision:

1. **Citation-grounded generation**: Every legal claim cites a specific case, statute, or regulation. Uncited assertions are prohibited.
2. **Jurisdictional awareness**: Legal rules vary by jurisdiction; Zen Legal explicitly identifies which jurisdiction's law it is applying and flags conflicts.
3. **Uncertainty disclosure**: The model expresses calibrated uncertainty about unsettled legal questions rather than stating contested doctrine as established law.
4. **Privilege protection**: Attorney-client privileged content is handled with enhanced access controls and never leaves the client environment.
5. **Professional responsibility compliance**: Outputs comply with ABA Model Rules and applicable state professional conduct rules.

## 2 Legal Domain Adaptation

### 2.1 Training Corpus

Zen Legal is fine-tuned on a 380-billion-token legal corpus:

Table 1: Legal training corpus composition

| Source | Tokens (B) | Coverage |
|---|---|---|
| Federal case law (PACER, CourtListener) | 68 | 1789–2025 |
| State case law (all 50 states) | 84 | Selective post-1970 |
| Federal statutory code (USC) | 8.4 | Current |
| Code of Federal Regulations (CFR) | 12 | Current |
| Law review articles (HeinOnline) | 48 | 1820–2025 |
| International law (UN, EU, OECD) | 28 | 1945–2025 |
| Legal contracts and agreements | 92 | Commercial |
| Bar exam materials and treatises | 14 | Current |
| Synthetic legal QA pairs | 26 | All jurisdictions |

### 2.2 Citation Grounding Architecture

Legal citations require hyper-precise attribution: case name, reporter, volume, page, pinpoint page, court, and year. Zen Legal's citation generation combines:

1. **Retrieval**: A dense retrieval index over 48 million legal documents retrieves candidate authorities.
2. **Verification**: Retrieved documents are verified against a citation registry that checks existence and accuracy of pinpoint cites.
3. **Shepardization**: Citations are checked against a legal validation service to flag overruled, distinguished, or limited precedents.
4. **Attribution**: The model generates text with inline citation markers that link to verified source documents.

Citation accuracy (citation exists and supports the stated proposition) reaches 96.8% on internal held-out evaluation by licensed attorneys.

## 3 Multi-Jurisdiction Reasoning

### 3.1 Jurisdiction Classification

Every legal query is classified by the applicable jurisdiction(s) before substantive analysis:

- Primary jurisdiction: The jurisdiction whose law governs the issue.
- Secondary jurisdictions: Related jurisdictions consulted for persuasive authority.
- Conflict detection: When jurisdictions differ, the model explicitly identifies the split.

$$P(j \mid q) = \text{softmax}(\mathbf{W}_j \cdot \text{ZenMoDE}(q)) \tag{1}$$

Jurisdiction classification accuracy: 97.4% for US federal/state classification, 94.2% for international jurisdiction identification.

### 3.2 Circuit Splits and Unsettled Law

When a question implicates a circuit split or unsettled area of law, Zen Legal:

1. Identifies the split (e.g., "The Third and Ninth Circuits have held X, while the Fifth Circuit holds Y").
2. States which rule applies in the user's jurisdiction.
3. Notes any Supreme Court certiorari activity or pending resolution.
4. Advises the user to verify current status given the dynamic nature of the area.

## 4 Contract Analysis

### 4.1 CUAD Benchmark Tasks

The Contract Understanding Atticus Dataset (CUAD) includes 41 clause types across commercial contracts. Zen Legal achieves the following results:

Table 2: CUAD contract clause extraction F1 (top 15 clause types)

| Clause Type | F1 | Recall |
|---|---|---|
| Governing Law | 0.972 | 0.981 |
| Limitation of Liability | 0.941 | 0.958 |
| Indemnification | 0.928 | 0.944 |
| Non-compete | 0.918 | 0.934 |
| Termination for Cause | 0.924 | 0.941 |
| Assignment | 0.931 | 0.948 |
| IP Ownership | 0.912 | 0.928 |
| Audit Rights | 0.904 | 0.921 |
| Change of Control | 0.896 | 0.914 |
| Warranty Duration | 0.884 | 0.902 |
| Renewal Term | 0.918 | 0.934 |
| Most Favored Nation | 0.872 | 0.891 |
| Exclusivity | 0.868 | 0.888 |
| Anti-assignment | 0.914 | 0.931 |
| Force Majeure | 0.894 | 0.912 |
| **CUAD Average (41 types)** | **0.923** | **0.938** |

### 4.2 Contract Risk Scoring

Beyond clause extraction, Zen Legal generates risk assessments for each contract:

- **Clause-level risk**: Each extracted clause is scored on a 1–5 risk scale relative to market standard terms.
- **Missing clause detection**: Identifies standard clauses absent from the agreement (e.g., no limitation of liability, no governing law clause).
- **Inconsistency detection**: Flags internally inconsistent provisions (e.g., conflicting notice periods in different sections).
- **Redline suggestions**: Proposes alternative language for high-risk clauses with explanation.

# 5 Legal Benchmark Results

## 5.1 LegalBench

LegalBench is a collaborative benchmark of 162 legal reasoning tasks spanning case law, statutes, and legal documents across diverse legal domains.

Table 3: LegalBench results by legal domain

| Domain | Tasks | Accuracy |
|---|---|---|
| Contract law | 28 | 84.2% |
| Constitutional law | 18 | 79.4% |
| Administrative law | 14 | 77.8% |
| Criminal law | 22 | 82.4% |
| Tort law | 16 | 83.8% |
| Statutory interpretation | 24 | 80.2% |
| International law | 12 | 76.8% |
| Procedural law | 28 | 83.4% |
| **LegalBench Overall** | **162** | **81.4%** |

## 5.2 Case Outcome Prediction

Table 4: Case outcome prediction accuracy by court level

| Court | Cases | Accuracy | AUC |
|---|---|---|---|
| U.S. Supreme Court | 8,412 | 78.4% | 0.841 |
| U.S. Circuit Courts | 142,841 | 74.8% | 0.812 |
| U.S. District Courts | 284,121 | 72.4% | 0.784 |
| State Supreme Courts | 48,412 | 71.8% | 0.778 |
| **Overall** | — | **74.2%** | **0.804** |

## 5.3 Legal NLP Tasks

Table 5: Legal NLP task results

| Task | Dataset | Metric | Score |
|---|---|---|---|
| Statute classification | EUR-Lex | Micro-F1 | 0.848 |
| Legal judgment prediction | ECHR | Accuracy | 81.4% |
| Obligation detection | ContractNLI | F1 | 0.872 |
| Legal argument mining | ECHRArguments | F1 | 0.814 |
| Regulatory compliance | PrivacyQA | F1 | 0.864 |
| Legal summarization | Multi-LexSum | ROUGE-L | 0.384 |

# 6 Legal Research Workflow

## 6.1 Memorandum of Law Generation

Zen Legal generates complete legal memoranda following standard structure:

1. **Question Presented**: Precise formulation of the legal question.

2. **Brief Answer**: Concise answer with legal conclusion and confidence.
3. **Statement of Facts**: Relevant factual background.
4. **Discussion**: IRAC (Issue, Rule, Application, Conclusion) analysis with full citations.
5. **Conclusion**: Summary of findings and recommended next steps.

Attorney evaluation of generated memoranda (blind comparison vs. associate-drafted):

Table 6: Attorney evaluation of AI-generated legal memoranda (N=150)

| Criterion | Zen Legal Score | Associate Avg |
|---|---|---|
| Legal accuracy | 4.24 / 5 | 4.18 / 5 |
| Citation completeness | 4.41 / 5 | 3.84 / 5 |
| Analytical clarity | 4.08 / 5 | 4.28 / 5 |
| Conclusion soundness | 4.18 / 5 | 4.22 / 5 |
| **Overall** | **4.23 / 5** | **4.13 / 5** |

# 7 Professional Responsibility Compliance

## 7.1 Unauthorized Practice of Law

Zen Legal explicitly frames all outputs as legal research assistance, not legal advice. The system:

- Appends standard disclaimers: "This output does not constitute legal advice and does not create an attorney-client relationship."
- Monitors for jurisdiction-specific UPL rules that constrain AI-generated legal content.
- For consumer-facing deployments, routes complex matters to licensed attorney review before delivery.

## 7.2 Competence and Supervision

Per ABA Model Rule 5.3, attorneys using AI tools must supervise the AI's work competently. Zen Legal supports this by:

- Providing explicit confidence scores on all legal conclusions.
- Flagging recently decided cases (<6 months) that may not be fully integrated in training data.
- Recommending verification steps for time-sensitive or high-stakes matters.

# 8 Conclusion

Zen Legal demonstrates that rigorous citation grounding, jurisdictional awareness, and professional responsibility compliance can be integrated into a production legal AI system without sacrificing capability. The 81.4% LegalBench score, 89.3% CUAD clause extraction, and 0.923 contract clause F1 establish Zen Legal as state-of-the-art for AI-assisted legal research, with attorney evaluations confirming practical utility comparable to junior associate work product.

## References


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